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Computer Science 

Class 10

FBISE

Chapter 1

Exercise Solution

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Computer Science 10th

Chapter no 1

Exercise: (Q/A)

Short Q/A
Qno1: What is the significance of Most Significant Bit(MSB) in signed binary numbers?
Significance of the Most Significant Bit (MSB) The MSB indicates whether a signed binary number is positive or negative. If the MSB is 0, the number is positive. If the MSB is 1, the number is negative.
Qno2: What is binary digit, and why it is significant in computer system?
Binary Digit and its Fundamental Nature A binary digit (bit) is the smallest data unit in computing that holds either a 0 or a 1. It is fundamental because digital electronic circuits naturally operate using two distinct physical states: ON (1) and OFF (0).
Qno3: Why are binary numbers more efficient for computer calculation than decimal numbers?
Efficiency of Binary Over Decimal Numbers Binary numbers are more efficient because they align directly with the two-state nature of electronic components (high and low voltage levels). This two-state design simplifies circuit complexity, minimizes processing errors, and drastically enhances calculation speed compared to the ten-state decimal system.
Qno4: How do ASCII and Unicode different in character representation?
Differences Between ASCII and Unicode ASCII: Uses 7 or 8 bits per character, restricting its capacity to representing standard English text, numbers, and basic punctuation (up to 256 unique symbols). Unicode: Supports up to 32 bits, enabling it to represent over 140,000 unique characters from all modern and historical world languages, technical symbols, and emojis.
Qno5: What is the importance of positional value in number systems? Give two examples.

Importance of Positional Value

Positional value is critical because it defines each digit's exact weight or value based on its location inside a number system.

Examples

Example 1 (Decimal): In the number 345, the positions correspond to hundreds, tens, and ones ($300 + 40 + 5 = 345$).

Example 2 (Binary): In the binary number 1010, the placement from right to left represents powers of two ($2^3, 2^2, 2^1, 2^0$), which equals $8 + 0 + 2 + 0 = 10$ in decimal.

Qno6: What is the process to convert a binary number into its hexadecimal?

Process to Convert Binary to Hexadecimal

Binary to hexadecimal conversion relies on grouping bits.

- Group binary digits into sets of four.
- Begin grouping from the right side.
- Add leading zeros to the left if needed.
- Convert each 4-bit group to its hex digit.
- **Example:** $(110101)_2 \gg (0011)_2 (0101)_2 \gg (35)_{16}$

Qno7: What is the purpose of machine code in computer operations ?

Purpose of Machine Code

Machine code consists of binary instructions (0s and 1s).

- The CPU executes it directly.
- No translation software is required.
- It directly controls hardware logic gates.

Qno8: Why coding scheme is used in computer? Give three reasons?

Why Coding Schemes are Used

Coding schemes map characters to unique binary codes. Three reasons:

- **Standardization:** Ensures uniform text interpretation worldwide.
- **Data Representation:** Allows computers to store non-numeric text.
- **Interoperability:** Enables different devices to exchange data accurately.

Qno9: State five differences between a process and a Thread.

Comparison between Process and Thread

Process	Thread
An executing instance of a program is called a process.	A thread is a subset of the process.
It has its own copy of the data segment of the parent process.	It has direct access to the data segment of its process.
Any change in the process does not affect other processes.	Any change in the thread may affect the behavior of the other threads of the process.
Processes run in separate memory spaces.	Threads run in shared memory spaces.
Process is controlled by the operating system.	Threads are controlled by programmer in a program.
Processes are independent.	Threads are dependent.

Qno10: What is memory management and how does it work in an operating system?

Memory management is the part of operating system that controls and manages the operation of main memory during the operation of computer.

Working

1. It allocates space to programs that are loaded in main memory for execution.
2. It keeps track of freed memory when a program is closed and updates the memory status.

Long Q/A

Qno1: Explain decimal, binary, octal and hexadecimal number systems with examples.

Decimal Number System

The decimal (also called Denary) number system is composed of the 10 symbols: 0, 1, 2, 3, 4, 5, 6,

7, 8 and 9. Using these symbols we can express any quantity. The decimal number system is called base 10 system because it has 10 digits. The decimal number system is a positional-value system, in which the value of a digit depends on its position in the number. For example, consider the decimal number 964.

- 9 represents 9 hundreds
- 6 represents 6 tens
- 4 represents 4 units

The 9 carries the most weight. It is called the **Most Significant Digit (MSD)**. The 4 carries the least weight and is called the **Least Significant Digit (LSD)**. Each digit of a number carries weight that can be expressed as powers of 10. The decimal point separates the positive powers of 10 from the negative powers.

$$\begin{aligned}(8745.215)_{10} &= 8 \cdot 10^3 + 7 \cdot 10^2 + 4 \cdot 10^1 + 5 \cdot 10^0 + 2 \cdot 10^{-1} + 1 \cdot 10^{-2} + 5 \cdot 10^{-3} \\ &= 8000 + 700 + 40 + 5 + 0.2 + 0.01 + 0.005 \\ &= 8745.215\end{aligned}$$

Binary Number System

In binary number system there are only two symbols or digits, 0 and 1. The base of binary number system is 2. All the statements made earlier about the decimal number system are equally applicable to the binary number system as well. The binary system is also a positional-value system, wherein each bit has its own value or weight expressed as power of 2.

$$\begin{aligned}(1011.101)_2 &= 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} \\ &= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125 \\ &= (11.625)_{10}\end{aligned}$$

The binary number $(1011.101)_2$ is represented above and its equivalent decimal value can be found by taking the sum of the products of each bit value (0 or 1) times its positional value. The left most bit carries the most weight and it is called the **Most Significant Bit (MSB)**. The positions to the right of the binary point are negative powers of 2. The right most bit carries the least weight and is referred to as the **least significant bit (LSB)**.

Octal Number System

The Octal number system has a base of 8, meaning it has eight possible digits; 0, 1, 2, 3, 4, 5, 6,

and 7. Thus, each digit of an octal number system can have any value from 0 to 7. The digit positions in an octal number system have weight with powers of 8. An octal number can be easily converted to its decimal equivalent by multiplying each digit by its positional weight.

For example

$$(628)_8 = 6 \cdot 8^2 + 2 \cdot 8^1 + 8 \cdot 8^0 = 384 + 56 + 2 = (442)_{10}$$

Also

$$(25.6)_8 = 2 \cdot 8^1 + 5 \cdot 8^0 + 6 \cdot 8^{-1} = 16 + 5 + 0.75 = (21.75)_{10}$$

Hexadecimal Number System

The hexadecimal number system uses base 16. Thus it has 16 possible digits. It uses the digits 0 - 9 and letters A, B, C, D, E and F as the 16 digit symbols. The table shows the relationship among hexadecimal and decimal numbers. It is important to remember that hexadecimal digits A through F are equivalent to the decimal values 10 through 15.

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

A hexadecimal number can be converted to its decimal equivalent by using the fact that each digit position has a weight that is power of 16. The LSD has a weight of ($16^0 = 1$) the next higher digit has a weight of ($16^1 = 16$), the next higher digit has a weight of ($16^2 = 256$), and so on.

The conversion process is shown below.

$$\begin{aligned}
 (3AF)_{16} &= 3 \cdot 16^2 + A \cdot 16^1 + F \cdot 16^0 \\
 &= 3 \cdot 16^2 + 10 \cdot 16^1 + 15 \cdot 16^0 \\
 &= 768 + 160 + 15 = (943)_{10}
 \end{aligned}$$

Qno2:What is 2's complement, and why it is used? Give examples.

.2's Complements

The binary numbers can be converted into the 2's complement with the help of a very simple technique.

- Take the 1's complement of the binary number
- Add 1 to the LSB (least significant bit)

Example 1: 2's complement of binary number 01010001

```

01010001 (Number)
10101110 (1's Complement)
+ 1 (Adding 1)
10101111 (2's Complement)

```

Example 2: 2's complement of binary number 10101110

```

10101110 (Number)
01010001 (1's Complement)
+ 1 (Adding 1)
01010010 (2's Complement)

```

Why 2's Compliments used?

In digital logic, 'Complements' are used to perform logical operations, especially the subtraction operation. Modern computers use 2's complement representation for several critical reasons:

Simplifies Hardware Design: It allows the CPU's Arithmetic Logic Unit (ALU) to perform subtraction using addition circuitry. The computer does not need a separate subtractor circuit; it just adds a negative number's 2's complement representation to a positive number ($A - B = A + (-B)$).

Single Representation for Zero: Unlike other systems (like Sign-Magnitude or 1's complement), 2's complement provides a unique representation for zero (00000000), eliminating the confusion of having both a "+0" and a "-0".

Clear Sign Identification: The Most Significant Bit (MSB) acts clearly as a sign bit: if the MSB is 0, the number is positive; if the MSB is 1, the number is negative

Qno3: Define the terms Overflow and Underflow. Explain Overflow in number system with one example.

Overflow and Underflow

In computers, overflow and underflow are problems that occur when calculations produce results that are too large or too small to be represented within the allowed number of bits. These

conditions are important because they can cause errors in programs and lead to unexpected behavior or bugs.

Underflow: Underflow happens when a calculation produces a result that is smaller than the minimum non-zero value that can be stored or represented with the available number of digits or bits in a system. This causes the system to be unable to properly show or handle the microscopic fractional value, typically forcing it to round down to zero.

Overflow: Overflow happens when a calculation produces a result that is larger than the maximum value that can be stored or represented with the available number of digits or bits in a system. This causes the system to be unable to properly show or handle the result.

Example: 8-bit Unsigned Integer Overflow

An 8-bit unsigned integer can represent values from 0 to 255. If user attempts to add 1 to 255, the result is 256, which cannot be represented with just 8 bits.

```
      11111111 (255 in binary)
+           1
-----
1-00000000 <<<< Overflow occurs (1 gets carried out of the 8 bits)
-----
      100000000 (256 in binary)
```

Qno4:What is the significance of ASCII and Unicode character coding schemes?

Common Coding Schemes (ASCII and Unicode)

Coding schemes are standards that computers use to convert and represent data in a format they can understand, such as binary digits (1's and 0's). Since computers cannot interpret human-readable characters like letters, numbers, and symbols directly, they need to translate this data into a machine-readable format. Different coding schemes are used for this purpose, among them ASCII and Unicode are common.

ASCII Coding Scheme

ASCII stands for **American Standard Code for Information Interchange**. It is a standard data coding scheme in computers. It assigns standard numeric values to alphabets, numerals,

punctuation marks and other characters used in computers.

On June 17, 1963 ASCII was approved as the American standard code. Initially ASCII used 7-bit codes of various combinations of 0's and 1's to represent 128 different characters ($2^7 = 128$). ASCII-7 could not be able to represent newly developed characters so it underwent further developments and revisions and ASCII-8 was developed. ASCII was extended to an 8-bit code that can represent ($2^8 = 256$) different characters. ASCII-8 became popular in 1981 when it was used first time in IBM's Personal Computer (PC) and soon it became industry standard for personal computers

Unicode Coding Scheme

Unicode(**Universal coding system**) is an international character-encoding system designed to facilitate the electronic exchange, processing, and display of written texts from diverse modern and classical languages. The Unicode Standard encompasses letters, digits, accents, punctuation marks, technical symbols for the world's major written languages, as well as emoji and other symbols, all using a uniform encoding scheme. The latest version of Unicode includes over 100,000 characters, Unicode assigns a unique number to each character that remains consistent across all systems that support Unicode.

Significance of ASCII and Unicode Coding Schemes

Data Representation: Computers use binary (0s and 1s) to represent all data. Coding schemes map human-readable characters to binary codes, making it possible for computers to store and manipulate text and other data.

Standardization: Coding schemes provide a consistent way to represent characters across different platforms and devices, ensuring compatibility and accurate data exchange.

Efficiency: They optimize the storage and processing of data by providing a fixed-length binary representation for each character.

Communication: They enable different computer systems to communicate with each other by using a common representation for characters and symbols.

Qno5:What is an operating system? Explain any five tasks/functions of OS.

Operating System

An operating system (OS) is essential software that manages a computer's memory, processes, hardware, and software, enabling user interaction. It loads into RAM when the computer is turned on and is necessary to run any programs. OS has evolved since the first computer generation and is present in all devices like desktops, tablets, and smartphones, with common examples being

Windows, Mac OS, Linux, iOS, and Android.

Main Tasks/Functions of an Operating System

Operating system performs the following main tasks or functions.

- Process Management
- Memory Management
- File Management
- Device Management
- Network Management
- Security management

Process Management

A process is a program in execution, and process management in an operating system allocates resources like CPU time to various processes in memory.

For example, with three processes (A, B, and C), each having different CPU cycles (A = 5 ms, B = 2 ms, C = 1 ms), the OS manages execution times. In Case 1 (ABC order), the total time is 6.67 ms, while in Case 2 (BCA order), it's 4.33 ms, showing that Case 2 is more efficient.

Memory Management

Memory management is the part of operating system that controls and manages the operation of main memory during the operation of computer. It allocates space to programs that are loaded in main memory for execution. It keeps track of freed memory when a program is closed and updates the memory status.

Example: In this example the OS is managing memory for two processes P0 and P1. P1 is being loaded (swap in) and P0 is being taken out (swap out) from the main memory (RAM).

File Management

File management is the part of operating system that manages files and folders on storage devices such as hard disk, USB flash drive and DVD. It allows computer user to perform operations such as creating, copying, moving, renaming, deleting, and searching files and folders. It also allows the user to perform read, write, open and close operations on files and folders.

Device Management

Device management is the part of operating system that controls input and output devices, such as keyboards, mouse, printers, and network interfaces. Efficient I/O management improves the performance of the computer.

Example: There are three programs A, B and C which are using the printer. Now the OS will decide which program to use the printer first. A queue will be set by the OS and each program will get the printer by its turn.

Network Management

Network management is the part of network operating system that monitors and manages the resources of a network. It allows to create user groups and assigns privileges to them. It shares the network resources among users and detects and fixes network problems.

Security management

Security management in an operating system ensures resources are used according to user privileges set by the system administrator. It creates user accounts, enforces security policies, controls access to resources, and protects the system from unauthorized access and malware.

Qno6: Explain any four types of operating systems.

Types of Operating Systems

The following are the important types of operating systems that are commonly used on various computer systems.

Batch Processing Operating System

A batch processing operating system groups similar tasks into batches and executes them sequentially. It is ideal for repetitive tasks, such as payroll processing or generating bank statements, where the same operation is applied to multiple users or jobs. While this system provides efficiency in handling large volumes of similar tasks, it may introduce delays as jobs are processed one after another.

Multitasking Operating System

A multitasking operating system enables multiple tasks to be performed concurrently on a single CPU. The system rapidly switches between tasks, creating the impression that all tasks are running simultaneously. This capability is essential for modern personal computers and mobile devices,

where users often run several programs at once.

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Time-sharing Operating System

A time-sharing operating system divides CPU time into short intervals, known as time slices, and allocates each program a time slice in turn. This method allows multiple users or programs to interact with the system simultaneously, giving the appearance that all are being processed at the same time. Time-sharing systems are commonly used in environments such as banks, universities, and large organizations that require multiple users to access the system concurrently.

Real-time Operating System (RTOS)

A real-time operating system is designed to process data and provide immediate responses. It is used in environments where timing is critical, such as medical equipment, traffic control systems, and industrial automation. Real-time systems must operate within strict time constraints to ensure that tasks are completed within a specified deadline, making them highly reliable and responsive.

Multiprocessor Operating System

A multiprocessor operating system controls the operation of multiple CPUs within a single computer system. By distributing tasks among several processors, the system can handle large amounts of data more efficiently and increase the overall processing speed. Multiprocessor systems are used in environments requiring high performance, such as data centers, servers, and scientific research applications.

Distributed Operating System

A distributed operating system manages a network of computers, allowing programs to run on multiple machines simultaneously. In a distributed system, the operating system distributes tasks across different computers in the network, balancing the load and providing fast execution of application software. Users do not need to know which machine is processing their tasks, as the distributed OS manages this transparently. This system is widely used in cloud computing and other networked environments.

Embedded Operating System

An embedded operating system is integrated into the hardware of a specific device, such as a microwave oven, TV, or camera. It is designed to perform specialized tasks and operates automatically when the device is turned on. Embedded operating systems are optimized for the specific functions of the device and provide reliable performance with minimal resource requirements. They are commonly used in consumer electronics and industrial machines.

Qno7:How operating system manages applications? Explain five states of a process with diagram.

How OS manages to run Applications?

Application programs run on top of operating systems by utilizing the resources and services provided by the operating system. The operating system serves as an intermediary between the hardware and the application programs, managing tasks such as memory allocation, process scheduling, and device management.

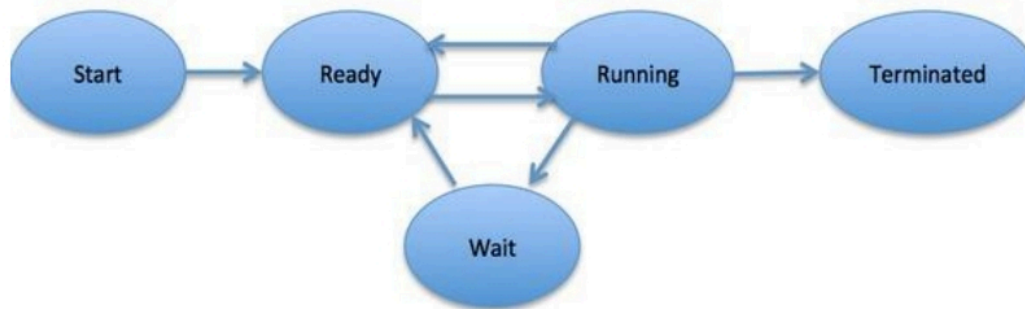
For example, when a user opens a word processing program such as Microsoft Word, the operating system allocates memory for the program to run, manages input/output operations such as reading and writing files, and schedules the program to run on the CPU. The operating system also provides access to system resources such as printers and network connections that the application program may need to use.

Process:

A process is a program in execution. For example, when we write a program in C or C++ and compile it, the compiler creates a binary code. The original code and Binary code, both are programs. When we actually run the binary code, it becomes a process.

Various States of a Process:

There are five states of a process which are start, ready, running, waiting and terminated



Start/New State: This is the first state of a process when it is created. Any new operation or service that is requested by a program for execution by the processor is known as new state of process.

Ready State: A process is said to be in ready state when it is ready for execution but it is waiting to be assigned to the processor by the operating system.

Running State: A process is said to be in running state when it is being executed by the processor. A process is assigned to a processor for execution by operating system.

Blocked State/Waiting State: A process is in blocked or waiting state when it is not under execution. It is waiting for a resource to become available.

Terminated State: A process is in terminated state when it completes its execution.

Qno8: Explain the role of Integrated Development Environments (IDEs) in the software development process, including specific features that enhance programming efficiency.

Integrated Development Environments (IDEs):

IDEs are comprehensive tools that combine a text editor, debugger, and compiler in one interface. Examples include IntelliJ IDEA, Eclipse, and Visual Studio.

Roles of IDE:

Writing Code: IDEs provide a platform for writing code with features like syntax highlighting, which colors the code according to its syntax, making it easier to read and understand. For example, keywords in a programming language might appear in blue, while comments are in green.

Code Completion: IDEs offer advanced code completion, where the software predicts and suggests the next part of the code you are typing. This feature helps in writing code faster and with fewer errors.

Debugging: IDEs come with integrated debuggers that allow programmers to run their code step-by-step, inspect variables, and understand the flow of execution to find and fix bugs (errors).

Error Checking and Syntax Highlighting: IDEs provide real-time error checking, underlining syntax errors and other issues as you type. This immediate feedback helps in quickly identifying and correcting mistakes.

Compiling Code: Compilers take the code written by programmers and translate it into executable machine code. They also perform various optimizations to make the code run more efficiently.

Project Management:

IDEs offer project management features that help organize code files, manage dependencies, and streamline the build process.

Qno9:What is software hosting? Explain On-premises and Cloud hosting. Also give two pros and cons of each.

Software Hosting

Software hosting refers to the process of deploying, managing, and providing access to software applications on servers. These servers can be located on-premises, in data centers, or in the cloud. The primary objective of software hosting is to make applications accessible to users over a network, typically the internet.

Some common types of Software Hosting are:

On-Premises Hosting:

On-premises hosting involves installing and running software on servers located within an organization's physical premises. This setup offers complete control over the hardware, software, and data security, making it ideal for organizations with strict compliance and security requirements. However, it comes with high upfront costs for purchasing and setting up the hardware, as well as ongoing maintenance and operational expenses.

Pros:

- You have complete control over hardware and data security.
- It is customizable to specific needs and compliance requirements.

Cons:

- It has high initial and ongoing costs.
- It requires in-house technical expertise. It has limited scalability.

Cloud Hosting:

Cloud hosting uses a network of virtual and physical servers to provide scalable and flexible resources on demand. Managed by cloud providers like AWS, Google Cloud, and Microsoft Azure, it offers high scalability, cost-efficiency with a pay-as-you-go model, and excellent reliability. Cloud hosting is ideal for applications with variable or unpredictable traffic. However, it depends on internet connectivity, and there may be concerns over data security and compliance.

Pros:

- It is highly scalable and flexible.
- It is cost-effective with pay-as-you-go pricing. It is reliable and supports global reach.

Cons:

- It is dependent on internet connectivity.
- There are potential data security and compliance concerns.